Type: MCQ

Q1. Which of the following is NOT a primary flight control surface? (0.5)

1. **Flaps
2. Elevators
3. Aileron
4. Rudder

Q2. The main function of the aircraft ailerons is to control: (0.5)

1. ** To control yaw motion
2. To control roll motion
3. To control pitch motion
4. To increase lift

Q3. Which type of landing gear is most commonly used in modern passenger aircraft? (0.5)

1. To control yaw motion
2. ** Tricycle landing gear
3. To reduce drag in cruise flight
4. To stabilize the aircraft in turbulence

Q4. The stall of an aircraft occurs when: (0.5)

1. The engine stops working
2. The landing gear fails to deploy
3. ** The angle of attack exceeds a critical value
4. The airspeed becomes supersonic

Q5. Which force always acts opposite to the direction of flight? (0.5)

1. Lift
2. Thrust
3. **Drag
4. Weight

Q6. The primary purpose of spoilers on an aircraft wing is to: (0.5)

1. ** Destroy lift and increase drag
2. Increase lift
3. Reduce drag
4. Control yaw

Q7. Which part of the aircraft provides stability in pitch? (0.5)

1. Vertical stabilizer

2. ** Horizontal stabilizer
3. Ailerons
4. Flaps

Q8. The aircraft Mach number is defined as the ratio of: (0.5)

1. Thrust to drag.
2. ** Speed of aircraft to speed of sound.
3. Lift to weight.
4. Airspeed to stall speed

Q9. What is the main purpose of wing flaps on an aircraft? (0.5)

1. To control yaw motion
2. ** To increase lift during takeoff and landing
3. To reduce drag in cruise flight
4. To stabilize the aircraft in turbulence

Q10. What is the purpose of the aircraft rudder? (0.5)

1. ** To control yaw motion
2. To control roll motion
3. To control pitch motion
4. To increase lift

Type: DES

Q11. Explain the difference between viscous and non-viscous flow regimes in aerodynamics. How do boundary layer effects influence drag and lift?. (4)

Ans:

**1. Viscous vs Non-viscous Flow Regimes**

- **Non-viscous (Inviscid) Flow:**
  - Assumes fluid has **no viscosity** ($\mu = 0$).
  - Governing equations neglect shear stresses → only pressure forces act.
  - Simplifies analysis → useful for potential flow theory, streamlines, lift predictions (e.g., thin airfoil theory).
  - Cannot capture drag due to skin friction or flow separation.
- **Viscous Flow:**
  - Accounts for **fluid viscosity**.
  - Shear stresses and velocity gradients are present.

- Captures **boundary layers**, **skin friction drag**, **wake formation**, and **flow separation**.
- Essential for accurate predictions of real aerodynamic performance.

**summary:** inviscid flow is an idealization (good for lift estimation in attached flow), while viscous flow represents reality, especially important for drag prediction.

**Boundary Layer Effects on Drag and Lift**

- **Boundary Layer (BL):** Thin region near a surface where velocity changes from zero (no-slip at wall) to free-stream.

**Influence on Drag:**

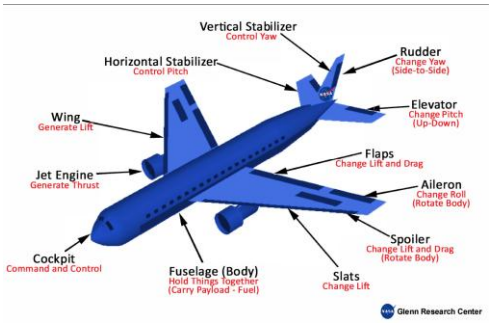
- **Skin Friction Drag:** Due to shear stress within the BL.
- **Pressure (Form) Drag:** If BL separates from the surface, flow wake forms → higher drag.
- Thicker/turbulent BL increases skin friction but delays separation → trade-off.

**Influence on Lift:**

- BL separation on an airfoil reduces suction peak on upper surface → lift decreases.
- At high angle of attack, BL separation causes stall → dramatic loss of lift.
- A well-managed BL (e.g., via turbulence transition or suction control) helps sustain attached flow, preserving lift.

Q12. With a neat sketch, explain the anatomy of an aircraft and describe the function of its major components. (3)

Ans:




**1. Fuselage**

- The **main body** of the aircraft.
- Houses cockpit (flight deck), passenger cabin, cargo, fuel systems, avionics.
- Provides attachment points for wings, tail, and landing gear.
- Function: Structural backbone; distributes loads and carries payload.

**2. Wing**

- Provides **lift** by creating pressure difference (airfoil section).

- Houses fuel tanks, flaps, ailerons, and sometimes landing gear.
- Function: Generates lift and enables roll control (via ailerons).

**3. Empennage (Tail Unit)**

- Consists of **horizontal stabilizer + elevator**, and **vertical stabilizer + rudder**.
- Horizontal stabilizer: provides pitch stability.
- Elevator: controls pitch (nose up/down).
- Vertical stabilizer: provides directional (yaw) stability.
- Rudder: controls yaw (left/right).

**4. Propulsion System**

- Engine (turbofan, turbojet, turboprop, or piston).
- Provides **thrust** to overcome drag.
- Nacelles/pylons integrate engines with the wing/fuselage.

**5. Landing Gear**

- Retractable/non-retractable wheels or skids.
- Provides ground mobility (takeoff, taxi, landing).
- Absorbs shock loads during landing.

**6. Control Surfaces**

- **Ailerons (on wings):** roll control.
- **Elevator (on tailplane):** pitch control.
- **Rudder (on fin):** yaw control.
- **Flaps/slats (on wings):** increase lift at low speeds (takeoff/landing).
- **Spoilers/airbrakes:** reduce lift, increase drag (descent, braking).

Q13. Compare lighter-than-air and heavier-than-air flight with suitable examples. (3)

Ans:

| **Aspect** | **Lighter-than-Air Flight (LTA)** | **Heavier-than-Air Flight (HTA)** |
|---|---|---|
| **Principle of Flight** | Achieves flight by **buoyancy** – displacement of air (Archimedes' principle). | Achieves flight by **aerodynamic lift** generated by wings/rotors. |
| **Main Lifting Medium** | Uses **lighter gases** than air (Helium, Hydrogen, Hot air). | Uses **airfoils** (fixed wings or rotary blades) to create pressure difference. |

| Aspect | Lighter-than-Air Flight (LTA) | Heavier-than-Air Flight (HTA) |
|---|---|---|
| **Examples** | Balloons, Airships (Zeppelins, Blimps). | Airplanes, Helicopters, Gliders, Drones. |
| **Thrust Requirement** | Not needed to stay aloft (only for movement/steering). | Continuous thrust required to balance drag and maintain lift. |
| **Speed and Range** | Slow, limited speed, not suitable for long-distance high-speed travel. | High speed, long range, suitable for transport, military, space launch. |
| **Stability and Control** | More susceptible to wind/weather, large size makes maneuvering harder. | Greater maneuverability, stability, and control using control surfaces. |
| **Applications** | Observation, advertising, leisure flights, early military reconnaissance. | Passenger transport, cargo, military aircraft, space access (rockets). |

Q14. Explain how the angle of attack influences the pressure distribution and aerodynamic forces on a wing. (3)

Ans:

**Definition**

- **Angle of attack (AoA):** the angle between the **chord line** of an airfoil and the **oncoming free-stream airflow**.
- It is a key parameter governing **lift and drag** on a wing.

**Influence on Pressure Distribution**

- At **low AoA**:
    - Flow remains attached.
    - Pressure on the **upper surface** is slightly lower than ambient (suction peak).
    - Pressure on the **lower surface** is slightly higher → small lift.
- At **moderate AoA**:
    - Suction on the upper surface increases significantly (larger negative pressure).
    - Pressure difference between upper and lower surfaces increases → **lift increases** nearly linearly with AoA (within linear region).
- At **high AoA (near critical AoA):**
    - Flow separation begins on the upper surface.
    - Suction peak reduces, pressure difference weakens.
    - Lift reaches **maximum (CLmax)**.
- At **beyond stall AoA:**
    - Flow separates massively.
    - Upper surface pressure recovery collapses.

- Lift decreases sharply, while drag increases → **stall condition**.

**Influence on Aerodynamic Forces**

- **Lift (L):** Increases linearly with AoA (in linear region), according to

$$C_L = C_{L0} + a\alpha$$

where $a$ is lift curve slope (~0.1 per degree for thin airfoils).

- **Drag (D):**
  - Induced drag increases with AoA (due to stronger vortices).
  - At stall, drag rises steeply due to separated wake.
- **Pitching Moment (M):**
  - AoA affects the pressure distribution and thus the moment around the aerodynamic center.
  - Generally, higher AoA increases nose-down pitching moment for cambered airfoils.

Q15. Define geometric altitude and geopotential altitude. Derive the relation between them. (3)

Ans:

**Definitions**

- **Geometric Altitude ($h$)**:
  The actual vertical distance of a point above the mean sea level, measured geometrically.
- **Geopotential Altitude ($h_g$)**:
  A fictitious altitude that accounts for the **variation of gravitational acceleration** $g$ with height. It is defined such that the potential energy per unit mass at height $h$ in the real atmosphere is the same as if gravity were constant at sea-level value $g_0$.

$$h_g = \frac{1}{g_0}\int_0^h g(h')\,dh'$$

**Gravitational Variation with Altitude**

Gravitational acceleration decreases with altitude as:

$$g(h) = g_0\left(\frac{R_E}{R_E + h}\right)^2$$

where:

- $R_E$ = mean radius of Earth (~6371 km)

- $g_0$= standard gravity at sea level (9.80665 m/s²).

## Derivation of Relation

- Start with definition:
- $$h_g = \frac{1}{g_0}\int_0^h g(h')\,dh'$$
- Substitute $g(h)$:
- $$h_g = \frac{1}{g_0}\int_0^h g_0(\frac{R_E}{R_E+h'})^2 dh'$$
$$h_g = \int_0^h (\frac{R_E}{R_E+h'})^2 dh'$$
- **Integration:**
- Let $u = R_E + h'$, so $du = dh'$.
- $$h_g = \int_{R_E}^{R_E+h} (\frac{R_E}{u})^2 du$$
$$h_g = R_E^2 \int_{R_E}^{R_E+h} \frac{1}{u^2} du$$
$$h_g = R_E^2[-\frac{1}{u}]_{R_E}^{R_E+h}$$
$$h_g = R_E^2(\frac{1}{R_E} - \frac{1}{R_E+h})$$
- **4. Final Relation**
- $$\boxed{h_g = \frac{R_E \cdot h}{R_E + h}}$$

## Approximation for Low Altitudes

- For $h \ll R_E$:
- $$h_g \approx h(1 - \frac{h}{R_E})$$
- Thus, at practical aircraft altitudes (<50 km), $h_g \approx h$, but for satellites and high-altitude calculations, the distinction is important.

Q16. Differentiate between Indicated Airspeed (IAS), Calibrated Airspeed (CAS), and True Airspeed (TAS). (3)

Ans:

| Type of Airspeed | Definition | Corrections Applied | Key Use |
|---|---|---|---|

| Type of Airspeed | Definition | Corrections Applied | Key Use |
|---|---|---|---|
| **Indicated Airspeed (IAS)** | The speed read directly from the airspeed indicator (ASI), which senses dynamic pressure from the pitot-static system. | No corrections applied — contains **instrument errors** and **position errors**. | Used by pilots for flying the aircraft (reference for stall speeds, maneuvering). |
| **Calibrated Airspeed (CAS)** | IAS corrected for **instrument error** and **position error** (caused by installation of pitot-static probe). | Corrections for instrument + position error. | More accurate than IAS; used in performance charts. |
| **True Airspeed (TAS)** | The actual speed of the aircraft relative to the undisturbed air mass through which it flies. | CAS further corrected for **air density** (altitude + temperature effects).<br>Formula: $TAS = CAS\sqrt{\frac{\rho_0}{\rho}}$ | Used in **navigation** and flight planning. |

Q17. An aircraft is flying at sea-level conditions where the air pressure is 101325 Pa and the temperature is 300 K. The velocity of the aircraft is 255 m/s. Calculate: The Mach number of the aircraft. (2)

Sol:

**Calculate Speed of Sound**

$$a = \sqrt{1.4 \times 287 \times 300}$$

First compute step by step:

- $1.4 \times 287 = 401.8$
- $401.8 \times 300 = 120540$
- $\sqrt{120540} \approx 347.2\ \text{m/s}$

So,

$$a \approx 347.2\ \text{m/s}$$

**Calculate Mach Number**

$$M = \frac{255}{347.2} \approx 0.735$$

Q18. Explain the difference between infinite wing and finite wing. (2)

Ans:

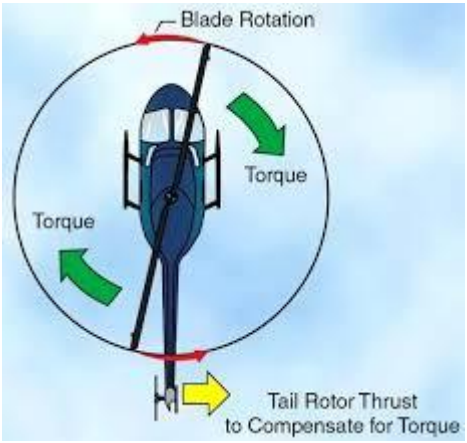
 An **infinite wing** is a theoretical wing with no tips, producing 2D flow, no induced drag, and a lift slope of $2\pi$ per radian.

 A **finite wing** has tips, produces wingtip vortices, experiences downwash, and therefore has lower lift and additional induced drag.

 Thus, real aircraft wings are finite, while infinite wing analysis is used for simplified airfoil theory.

Q19. With a neat sketch, explain any two functions of the main rotor and tail rotor in a helicopter. (2)

Ans:




 The **main rotor** generates lift and thrust (collective & cyclic control) and controls pitch and roll.

 The **tail rotor** counters main rotor torque and provides yaw control.